

**BACHELOR OF COMPUTER
APPLICATIONS (BCA) (REVISED)**

Term-End Examination

December, 2022

BCS-53 : WEB PROGRAMMING

Time : 2 Hours

Maximum Marks : 50

(Weightage : 75%)

Note : *Question No. 1 is compulsory and carries 20 marks. Attempt any **three** questions from question no. 2 to question no. 5.*

1. (a) Write a JavaScript and related HTML code which changes the content of a paragraph having an id "IGNOU" from "BCA Examination" to the text "The out of distance education". 4
- (b) Explain the box model of HTML with the help of a diagram. 3

- (c) Create an XML document having two records with the structure : 3
- **Name** which have first name and last name of a student.
 - **Programme** which is name of the programme of the student.
- (d) What is a cookie ? Why are the cookies used ? Write the code that can create a new cookie. 4
- (e) Write a JSP script that prints numbers 1 to 5 on a separate HTML paragraph. 4
- (f) List the advantages of JSP over Servlet. 2
2. (a) Explain the features of Web 2.0. Explain any *three* technologies of Web 2.0. 5
- (b) What is CSS ? Explain how CSS can be used for the following : 5
- (i) Change the background colour of a page to red.
 - (ii) Change the font size of a paragraph to 17 points.
 - (iii) Changes colour of a paragraph.

3. (a) What is a DTD in the context of an XML document ? Given the following DTD : 5

```
<!ELEMENT University (Name, Address)>
```

```
<!ELEMENT Name (# PCDATA)>
```

```
<!ELEMENT Address (Location, Pincode)>
```

```
<!ELEMENT Location (# PCDATA)>
```

```
<!ELEMENT Pincode (# PCDATA)>
```

Create a document having two records, which are valid as per the DTD given above.

- (b) Write the code in WML that creates the following table : 5

Course Name	Teacher
Web Programming	ABC
DBMS	XYZ

4. (a) Differentiate between the following : 6

- (i) GET and POST methods
- (ii) HTTP server and Web container
- (iii) Action elements and JSP implicit objects

- (b) What is an exception ? What are the different types of errors in JSP ? Give *one* example of each. 4

5. Explain the following with the help of an example :

$$2\frac{1}{2} \times 4 = 10$$

- (a) DriverManager class
- (b) PreparedStatement() method
- (c) ResultSet object
- (d) MVC architecture

**BACHELOR OF COMPUTER
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Term-End Examination

June, 2023

BCS-053 : WEB PROGRAMMING

Time : 2 Hours

Maximum Marks : 50

(Weightage : 75%)

Note : Question No. 1 is compulsory and carries 20 marks. Attempt any **three** questions from question no. 2 to question no. 5.

1. (a) Consider the following information : 3+3

Student Name		Programme Code
First name	Last Name	
Ravi	Kumar	BCA
Ananth	Malik	MCA

- (i) Create an XML document for the data given in the table above. Use appropriate tags.
- (ii) Create the DTD to validate the XML document created in part (i).

- (b) A database table consists of the following fields :

Programme (ProgrammeCode, Title, Duration)

Assuming that connection has been established with the database and a statement object “stmt” has been instantiated, write the code fragment to get a ResultSet object containing the list of all those programmes, whose duration is more than 1 year. 6

- (c) Write a JavaScript script that changes the colour of paragraph, whose id is “IGNOU”, to green. 5

- (d) Explain the GET method of HTTP with the help of an example. 3

2. (a) Explain the MVC architecture with the help of a diagram. 5

- (b) What are implicit objects in the context of JSP ? Explain any *two* implicit objects of JSP. 5

3. (a) What is the need of session management in the context of HTTP ? Define the terms—session and state in this context. Explain with the help of .JSP script, how session object can be used to create a session. 7
- (b) What is include directive in JSP ? Explain its use with the help of an example. 3
4. (a) Explain the following technologies in the context of Web 2.0 : 6
- (i) Widgets
 - (ii) Blogging
 - (iii) Podcasting
- (b) Write HTML code to create a form as shown below : 4

Student Name :							
Programme Code :	<table border="1"><tr><td>BCA</td><td>▼</td></tr><tr><td>BCA</td><td></td></tr><tr><td>CIT</td><td></td></tr></table>	BCA	▼	BCA		CIT	
BCA	▼						
BCA							
CIT							
SUBMIT							

5. (a) What are the advantages of using CSS ?
What is an external style sheet ? How is it
linked to the related HTML document ? 5
- (b) Explain the following WML elements with
the help of an example : 3+2
- (i) WML Tables
 - (ii) <anchor> element

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***Note : Question No. 1 is compulsory. Attempt any
three questions from the rest.***

1. (a) What is WAP ? How does WAP model work ? Explain it with diagram. 5
- (b) Explain the uses of any *five* HTML elements which are used in form design. 5
- (c) What is Document Object Model (DOM) ? Draw the node three of an HTML DOM. 3
- (d) What is AJAX ? List out the differences between AJAX and JavaScript. 4

- (e) What is CSS ? Explain embedded style sheets with HTML example. 3
2. (a) What is DTD ? Explain its role in XML. 5
- (b) What is the need of server side scripting ? How is it different from client side scripting ? 5
3. (a) What are cookies ? How do you create and set the life of cookie in a JSP program ? 5
- (b) Design an WML program to display username and password. Make the necessary assumptions. 3
- (c) Why do we use GET and POST methods of HTTP protocol ? 2
4. (a) Differentiate between the following : 4
- (i) Include directive and Include action
- (ii) Model versus view in the context of MVC architecture.
- (b) Explain the following WML elements : 3
- (i) <optgroup>
- (ii) <setvar>

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- (c) Explain the Box model of HTML with the help of a diagram. 3
5. (a) Write a JSP program to fetch the record of the student having attributes such as name, programme, course code and grade from the table stored in database. 6
- (b) Explain the method getElementById () in JavaScript with the help of an example. 4

No. of Printed Pages : 3

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Term-End Examination

June, 2024

BCS-053 : WEB PROGRAMMING

Time : 2 Hours

Maximum Marks : 50

Note : *Question No. 1 is compulsory. Attempt any
three questions from the rest.*

1. (a) Write a JSP program which will demonstrate the use of <jsp:forward> and <jsp:param> actions. 5
- (b) What is safe method ? Is Get method a safe method ? Give reason in support of your answer. 3

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- (c) Explain how errors are handled at application level in JSP programs, with the help of a program fragment. 4
 - (d) Why do we need XML ? Write advantages of XML over HTML. 5
 - (e) What is web container ? Name any *three* web containers. 3
2. (a) What are JSP Implicit objects ? Explain the use of any two implicit objects. 4
- (b) Differentiate between the following : 6
- (i) Static web pages and dynamic web pages.
 - (ii) 2-Tier architecture and 3-Tier architecture.
3. (a) Explain WML Navigation Elements with the help of WML programs. 5
- (b) Write a HTML program using <details> and <summary> tag. Also explain both the tags. 5
4. Explain the following with the help of an example/diagram, if needed : $4 \times 2\frac{1}{2} = 10$
- (a) WAP protocol stack

- (b) Deployment Descriptor
 - (c) XML XSLT
 - (d) JSP Scriptlets
5. (a) Explain the features of the following web technologies : 6
- (i) Mashups
 - (ii) Rich Internet Applications
 - (iii) Web services
- (b) Write a JSP program which may print the series of odd numbers from 1 to 100. Also find the sum of these numbers. 4

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Weightage : 75%

Note : *Question No. 1 is compulsory and carries 20 marks. Attempt any **three** questions from question no. 2 to question no. 5.*

1. (a) What is WAP ? Write WML program that displays your name, enrollment no. and course name in centre of the page. 4

- (b) Design an HTML form which should allow to enter data. (**Hint** : Make use of password field, text field, check boxes, radio and submit button.) 4
- (c) Write a JSP scriptlet that display the current date and time. 3
- (d) Explain the uses of XML CSS and XML XSLT. 3
- (e) Write a JavaScript program to find the largest among three numbers. 4
- (f) What is the need of JSP implicit object ? 2
2. (a) What is CSS ? How do you include comments in CSS ? Write CSS code for including a background image without repeating in a page. 5

- (b) What is session management in JSP ?
What are the different session tracking methods ? Explain any **one** of them. 5
3. (a) Differentiate between the following : 4
- (i) Server side and Client side scripting
 - (ii) Include directive and JSP include
- (b) Explain the use of deployment descriptor in JSP. Write code fragment for “File Not Found Error 404” using deployment descriptor. 4
- (c) Explain the Rich Internet Applications. 2
4. (a) Explain the steps involved in connecting to database using JDBC. 5
- (b) Explain the image and navigational elements of WML with the help of an example. 5

5. Explain any *four* of the following with the help

of an example :

$$4 \times 2\frac{1}{2} = 10$$

(a) CreateStatement() method

(b) <jsp:UseBean>

(c) Role of web container

(d) MIME

(e) <MENU> tag

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Term-End Examination

June, 2025

BCS-053 : WEB PROGRAMMING

Time : 2 Hours

Maximum Marks : 50

Weightage : 75%

Note : *Question No. 1 is compulsory and carries
20 marks. Attempt any **three** questions
from question nos. 2 to 5.*

1. (a) What is the purpose of <error-page>
element in JSP page ? Explain. 4

- (b) What is WML ? Explain any *two* WML elements that can be used for data inputs. 4
 - (c) What is DOM ? Draw the node tree of an HTML DOM. 4
 - (d) Write a JSP scriptlet that displays the area of a triangle. 4
 - (e) Explain the following tags with HTML code : 4
 - (i) <ASIDE> tag
 - (ii) <MATHML> tag
 - (iii) <BDI> tag
 - (iv) <NAV> tag
2. (a) Differentiate between the following : 6
- (i) executeUpdate() and executeQuery() method
 - (ii) Web container and HTTP server
 - (iii) Ajax and JavaScript

- (b) What is DTD ? Explain its role in XML. 4
3. (a) Write a JSP program which will demonstrate the use of < jsp : plug in > action element. 5
- (b) Why do we use GET and POST methods of HTTP protocol ? Explain with example code. 5
4. (a) Explain the use of the following in XML : 5
- (i) XML schemas
- (ii) XML namespaces
- (b) Explain the *two* attributes of page directive which are included in exception handling in JSP page. 5

5. Explain any *four* of the following terms with the help of a diagram or an example :

$$4 \times 2 \frac{1}{2} = 10$$

- (a) JDBC-ODBC Bridge
- (b) Inline style sheet
- (c) Taglib directive
- (d) Mashups
- (e) 3-tier architecture

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No. of Printed Pages : 5

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Time : 2 Hours

Maximum Marks : 50

Weightage : 70%

Note : (i) *Question No. 1 is compulsory and carries 20 marks.*

(ii) *Attempt any **three** questions from question no. 2 to question no. 5.*

1. (a) Write HTML code to create a webpage as shown below : 5

Student Detail

Name :

Password :

Course Code :

BCS-053

BCS-011

BCS-054

GO

- (b) What is the use of CSS ? Explain with the help of an example. 3
- (c) Explain the page directive element of JSP with syntax. 2
- (d) Explain the use of `getElementById( )` method embedded in HTML source code. 2

- (e) What is JDBC ? Write a JSP program to insert a record in the student table of database. Make and state suitable assumption, if any. 5
- (f) What is a cookie ? Write command to create a new cookie having a name 'txtcookie'. Also write a command to set the life of this cookie to 5 minutes. 3
2. (a) What is WML ? Design WML program to display the details of a student having attributes as enrolment number, name, program code and course code. 5
- (b) Explain the JSP life with the help of a diagram. 5
3. (a) Explain the role of each tier in 3-tier architecture of web application. 3

- (b) Write a JSP program to print the current date and time. 3
- (c) Differentiate between the following : 4
- (i) <jsp : include> and <jsp : forward>
 - (ii) Get and post method
4. (a) What is session management in JSP ?
Explain the different session tracking methods. 5
- (b) What is meant by Document Object Model in the context of HTML ?
Write a program using JavaScript and HTML that changes a paragraph text from 'IGNOU' to 'Welcome in IGNOU'. 5

5. Explain any *four* of the following with the help of an example/diagram, if needed :

$$4 \times 2\frac{1}{2} = 10$$

- (a) WML components
- (b) XML scheme
- (c) JDBC-ODBC Bridge
- (d) Deployment descriptor
- (e) Uses of <detail> and <summary> tags

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